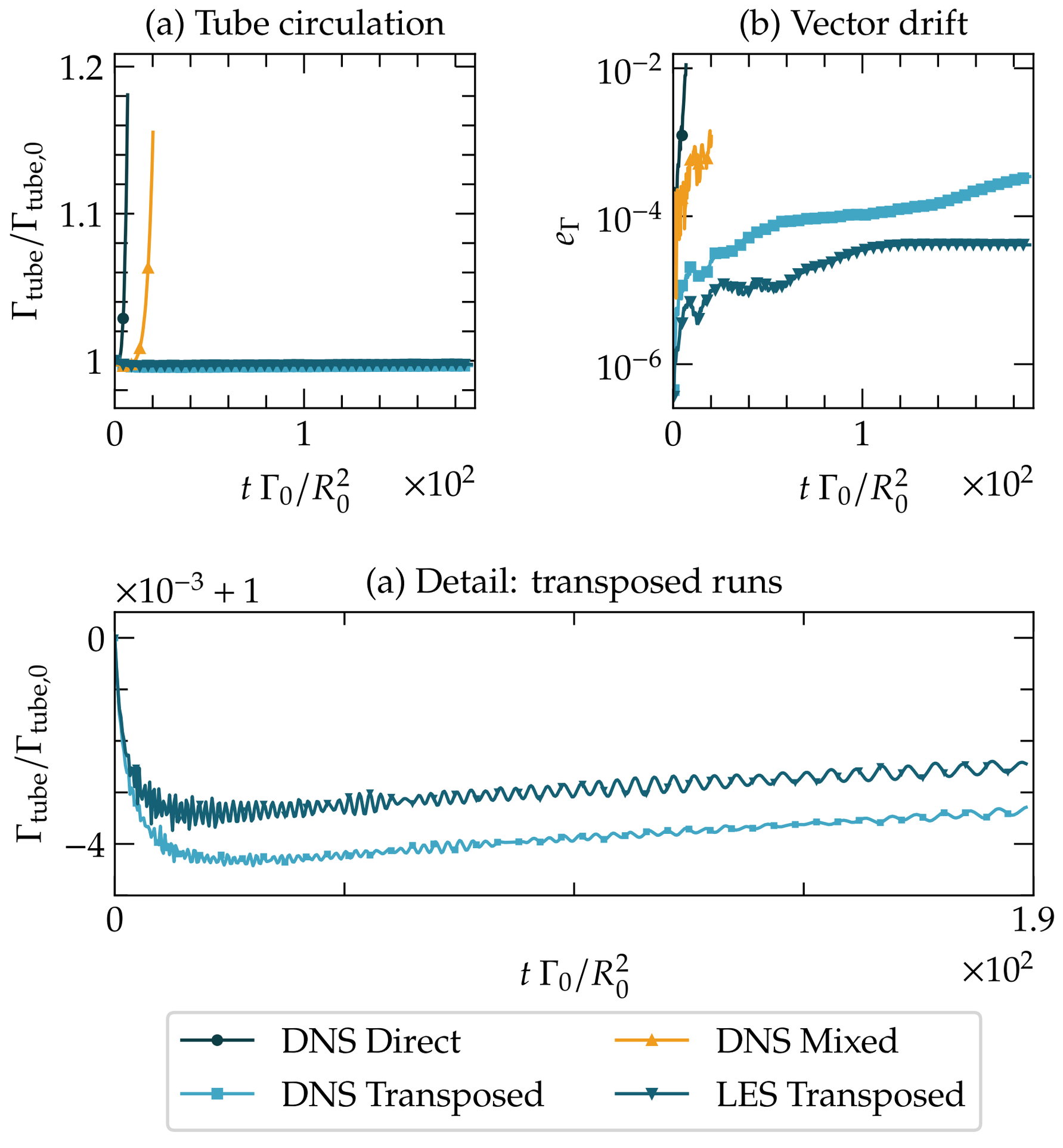

(a) Tube circulation
1.2
1.1
1
$\Gamma_{\text{tube}}/\Gamma_{\text{tube},0}$
0
1
$t\,\Gamma_0/R_0^2$
$\times 10^2$
(b) Vector drift
10^{-2}
10^{-4}
10^{-6}
e_Γ
0
1
$t\,\Gamma_0/R_0^2$
$\times 10^2$
(a) Detail: transposed runs
$\times 10^{-3} + 1$
0
−4
$\Gamma_{\text{tube}}/\Gamma_{\text{tube},0}$
0
1.9
$t\,\Gamma_0/R_0^2$
$\times 10^2$
DNS Direct
DNS Mixed
DNS Transposed
LES Transposed